Discrete Probability Distributions: Binomial, Poisson & Geometric

Statistical Data Analysis & Visualization Laboratory

Semester III • Lab Program 2

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Lab Program 2

Module	Module 2
Course Outcome	CO2
Aim	To compute, fit, and visualize discrete probability distributions and calculate their expectations, moments, and variance with real-world ML-relevant examples.

Introduction

- This lab introduces **three important discrete probability distributions** using Python:
  1. **Binomial Distribution** – number of successes in a fixed number of trials.
  2. **Poisson Distribution** – number of events occurring in a fixed interval.
  3. **Geometric Distribution** – number of trials required to get the first success.
- These concepts are useful in **Machine Learning, classification, anomaly detection, system reliability, NLP, network traffic analysis, and model evaluation**.

Learning Objectives

- Compute PMF and CDF of Binomial, Poisson, and Geometric distributions
- Verify theoretical mean and variance from simulated data
- Compute mathematical expectation, second moment, and variance
- Visualize and compare distributions across different parameters

Theory

Distribution	PMF	Mean	Variance	ML Application
Binomial(n,p)	$C(n,k)\ p^k(1-p)^{n-k}$	np	$np(1-p)$	Binary classification trials
Poisson(λ)	$e^{-\lambda}\ \lambda^k / k!$	λ	λ	Rare events, count data
Geometric(p)	$(1-p)^{k-1}\ p$	$1/p$	$(1-p)/p^2$	First success in repeated trials

Python Implementation

1. Import the libraries

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

- Numpy - Used for numerical operations and random data generation.
- Matplotlib.pyplot - Used to visualize the probability distributions.
- Scipy - This is particularly important. SciPy already provides functions for distributions such as: stats.binom ; stats.poisson ; stats.geom
- So we don't have to manually calculate every probability.

2. Set the random seed

```
np.random.seed(42)
```

- This makes the random simulation **reproducible**.
- If you run the program multiple times with the same seed, they will obtain the same random simulation results.

PART 1 – BINOMIAL DISTRIBUTION

3. Define the Binomial distribution

```
n, p = 20, 0.35
binom = stats.binom(n, p)
x = np.arange(0, n+1)
```

Here: **n = 20 & p = 0.35**
The distribution is:

$$X \sim Binomial(20, 0.35)$$

What does this mean?
We perform 20 independent trials, and each trial has: **P(success)=0.35**
The random variable **X** represents the number of successes among the **20** trials.

ML example

Suppose an ML model classifies 20 images.

Assume the probability that the model correctly classifies an image is:

$$p = 0.35$$

Then:

$$X = \text{number of correctly classified images}$$

can be modeled as:

$$X \sim Binomial(20, 0.35)$$

This is a simple example. In real ML evaluation, the probability of success may not be constant across observations, but the example is useful for understanding the distribution.

4. Mean of Binomial distribution

```
print(f' Mean = {binom.mean():.4f} (Theory: np={n*p})')
```

ML interpretation

If we classify 20 samples and the success probability is 0.35, we expect approximately:

7 successful classifications per 20 trials.

Remember:

> Expected value does not mean that every group of 20 samples will contain exactly 7 successes.

It means that over many repetitions, the average will approach 7.

5. Variance - `binom.var()`

Variance tells us about the **spread or variability** of the number of successes.

6. Second moment $E[X^2]$ – `binom.moment(2)`

Why are moments useful in ML?

Moments help describe a probability distribution.

For example:

- First moment → Mean
- Second moment → relates to variance
- Third standardized moment → Skewness
- Higher moments → help describe distribution shape

These ideas are important in **statistical modeling and feature/distribution analysis.**

7. Skewness - `binom.stats("s")`

– Skewness tells us about the asymmetry of a distribution.

For a Binomial distribution:

$$Skewness = \frac{1 - 2p}{\sqrt{np(1-p)}}$$

Since:

$$p = 0.35 < 0.5$$

the distribution is slightly **positively skewed**.

In simple words:

> The probability distribution has a somewhat longer tail toward larger values.

8. CDF – Probability of at most 10 successes

`binom.cdf(10)`

CDF means **Cumulative Distribution Function.**

It calculates:

$$P(X \leq 10)$$

In this experiment:

> What is the probability of getting **10 or fewer successes out of 20 trials?**

This is useful in ML when we want probabilities such as:

> What is the probability that the number of correct predictions is at most a particular threshold?

9. PMF – Probability of exactly 7 successes

`binom.pmf(7)`

PMF means **Probability Mass Function**.

It calculates:

$$P(X = 7)$$

In our ML example:

> What is the probability that exactly 7 out of 20 classifications are successful?

The Binomial PMF is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

PART 2 – POISSON DISTRIBUTION

```
for lam in [1, 3, 5, 10]:
    d = stats.poisson(lam)
    sim = np.random.poisson(lam, 5000)
```

Here the parameter is:

$$\lambda$$

called the **average rate** or **expected number of events**.

What does Poisson distribution represent?

Poisson is used to model the:

> **Number of times an event occurs during a fixed interval.**

Examples:

- Number of customers arriving in one minute
- Number of network packets arriving per second
- Number of server requests per minute
- Number of errors in a software system per hour
- Number of defects in a manufactured product
- Number of anomalies detected per hour

ML example: Anomaly detection

Suppose an ML-based monitoring system observes:

> On average, 5 anomalies occur every hour.

Then we can model:

$$X \sim Poisson(\lambda = 5)$$

Here:

$$\lambda = 5$$

means the expected number of anomalies per hour is 5.

- Theoretical mean - `d.mean()`

For a Poisson distribution:

$$E[X] = \lambda$$

λ	Mean
1	1
3	3
5	5
10	10

- Simulation - `sim = np.random.poisson(lam, 5000)`
- This generates **5,000 random observations** from the Poisson distribution.
- `sim.mean()` – calculates the average of the simulated observations.
- The simulation mean will not necessarily be exactly 5, but with many samples it should be close.
- This demonstrates the connection between:

Theoretical probability ↔ Simulated data

PART 3 – GEOMETRIC DISTRIBUTION

```
p_geo = 0.3
geo = stats.geom(p_geo)
```

The Geometric distribution answers:

How many trials do we need before we get the first success?

ML example

Suppose an ML model is used to generate candidate solutions.

Assume the probability that any generated candidate satisfies the required quality criterion is:

$$p = 0.3$$

We can ask:

> How many attempts will we need before obtaining the first successful candidate?

This can be modeled using a Geometric distribution.

Other examples:

- Number of attempts before successful model prediction
- Number of customer interactions before conversion
- Number of tests before detecting a fault
- Number of transmissions before successful packet delivery

- Mean of Geometric distribution

`geo.mean()`

For SciPy's `stats.geom`, X represents the **trial number on which the first success occurs**, so:

$$\boxed{E[X] = \frac{1}{p}}$$

For:

$$p = 0.3$$

we get:

$$E[X] = \frac{1}{0.3} = 3.3333$$

Therefore, on average, we need about **3.33 trials to obtain the first success**.

- Probability of success on the first trial

`geo.pmf(1)`

This calculates:

$$P(X = 1)$$

For a Geometric distribution:

$$P(X = k) = (1 - p)^{k-1}p$$

For $k = 1$:

$$P(X = 1) = p$$

Therefore:

$$P(X = 1) = 0.3$$

So there is a **30% probability that the first attempt itself is successful.**

- Probability that first success takes more than 5 trials

```
1 - geo.cdf(5)
```

The CDF gives:

$$P(X \leq 5)$$

Therefore:

$$1 - P(X \leq 5)$$

gives:

$$P(X > 5)$$

In simple words:

> What is the probability that we still haven't obtained a success after 5 attempts?

For $p = 0.3$:

$$P(X > 5) = (1 - 0.3)^5$$

$$= 0.7^5$$

$$\approx 0.1681$$

So there is approximately a:

$$\boxed{16.81\%}$$

chance that the first success requires more than five trials.

Important comparison

- The three distributions answer three different questions:

Distribution	Main Question	ML Example
Binomial	How many successes occur in n trials?	Number of correct predictions out of 20
Poisson	How many events occur in a fixed interval?	Number of anomalies per hour
Geometric	How many trials until the first success?	Attempts until first successful candidate

A simple way to remember them:
Binomial → Count successes
Poisson → Count events
Geometric → Count attempts

Connection to Machine Learning

- These distributions are not just mathematical exercises.
- They help you understand the probabilistic foundations of ML.

1. Binomial → Classification

  - Suppose: 100 independent predictions
  - Probability of correct prediction = 0.8
- We can ask: What is the probability of getting exactly 85 correct predictions?
- That's a Binomial-type question.

2. Poisson → Anomaly/Event Counts

- Suppose a cyber security system observes:
  - Average = 4 suspicious events/hour
- We can ask: What is the probability of observing 8 suspicious events in the next hour?
- That's a Poisson-type question.

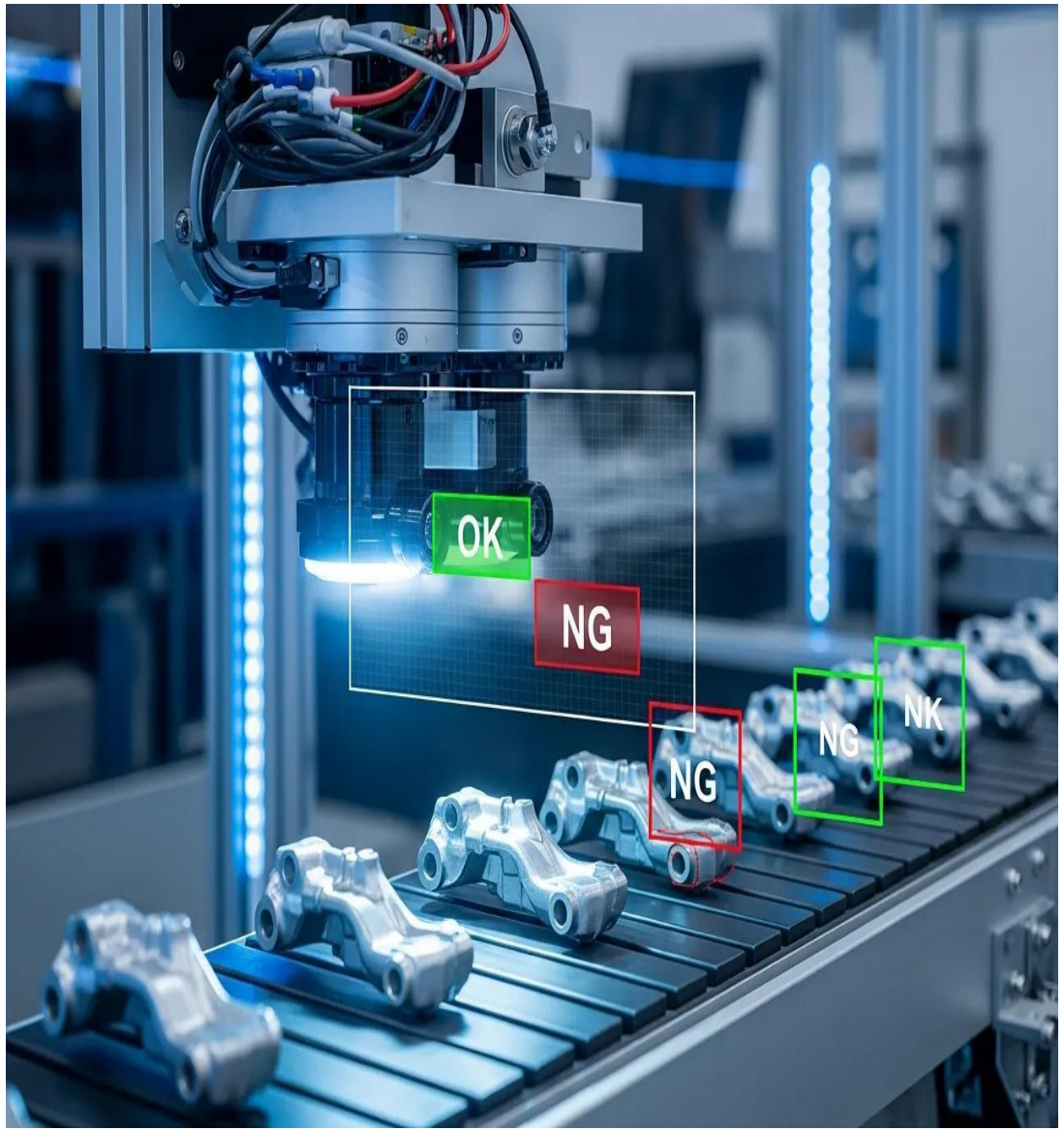
3. Geometric → Repeated Attempts

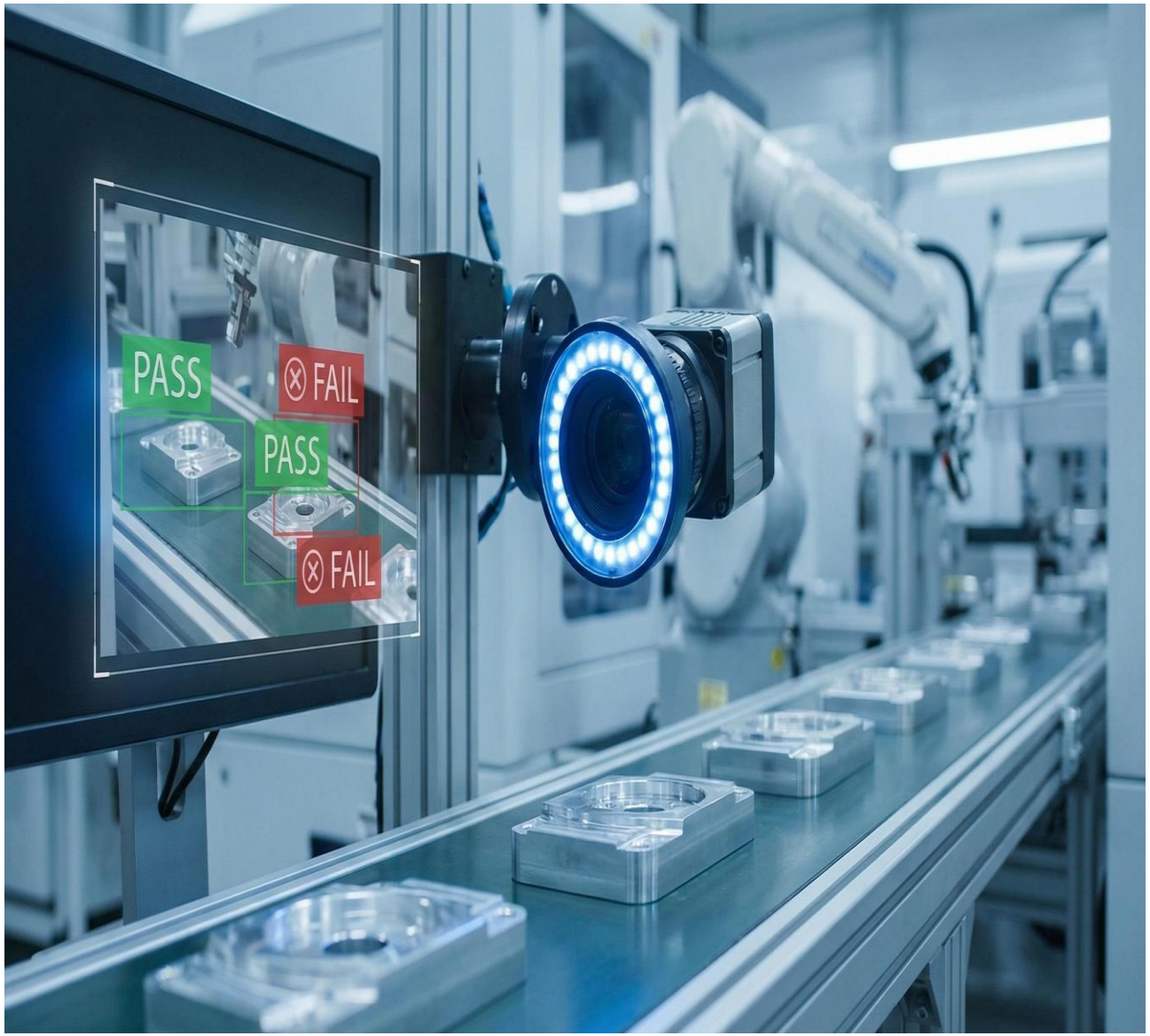
- Suppose an optimization algorithm has a 20% probability of finding an acceptable solution in each independent attempt.
- We can ask: What is the probability that the first acceptable solution occurs on the 5th attempt ?
- That's a Geometric-type question.

Industry use cases for the Discrete Probability Distributions

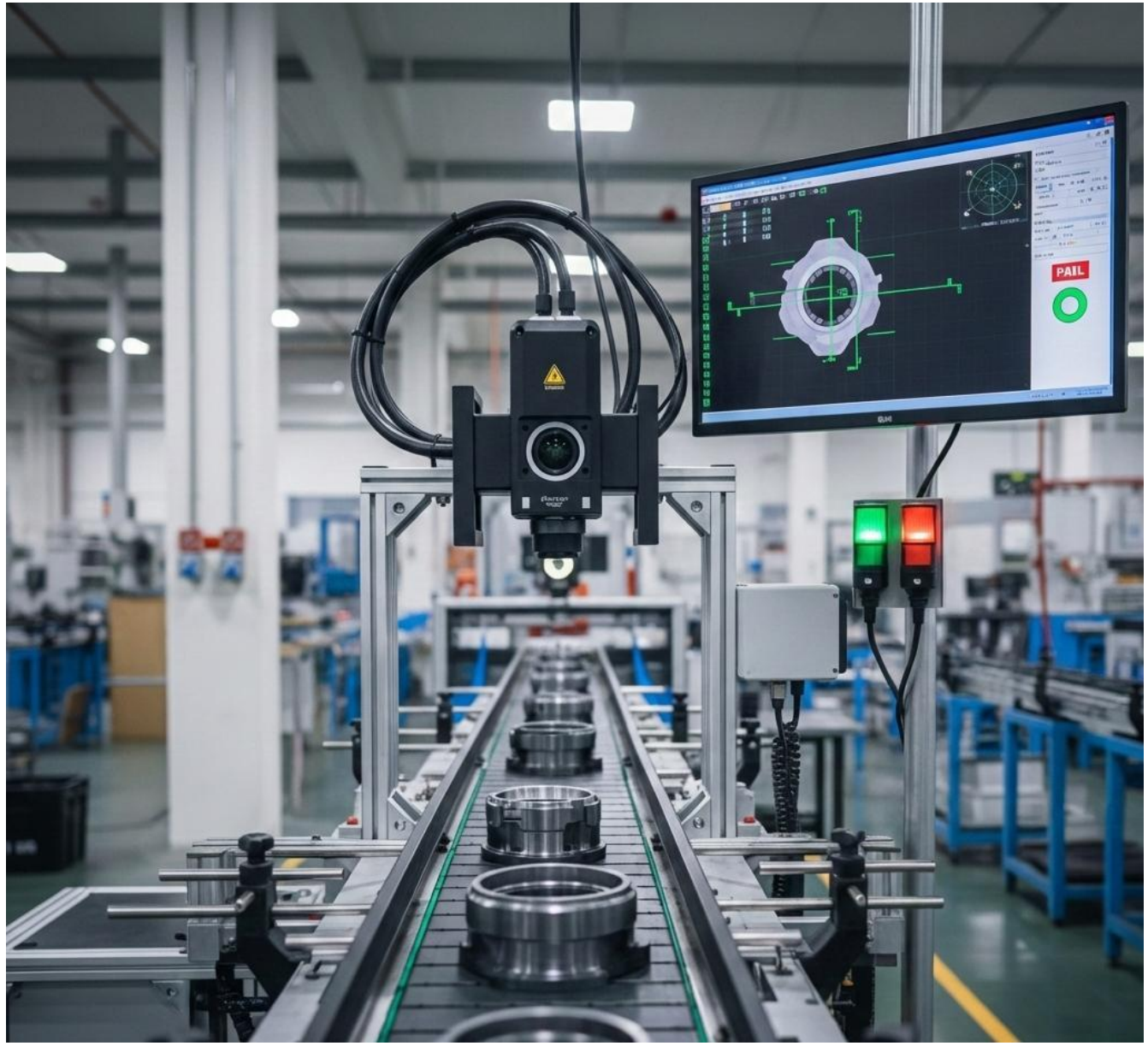
1. **Manufacturing Quality Control** — analyze defective products on a production line.
2. **Cloud Operations and Cybersecurity Monitoring** — analyze server requests and security incidents.

Use case 1: Manufacturing quality control






Visual Inspection
Trigger
Seting
camera
Last Fails or Info:
Inspection View
result:ok


FAIL


PASS
PASS
PASS
PASS
PASS
PASS
DEFECT: CRACK
DEFECT: CRACK
SPEED: 1200 PPM

Industry problem statement

A manufacturing company uses an AI-based computer vision system to inspect electronic components or manufactured products. Each product is classified as either defective or non-defective.

The quality-control engineer wants to answer:

- What is the probability of finding a particular number of defective products in a batch?
- How many defects should the company expect?
- How does the defect probability affect the distribution?
- How many inspections might be needed before the first defect is detected?

Sample dataset

- Suppose the inspection system estimates a defect probability of p=0.05, based on previous production data.
- There are 200 inspected products and 11 observed defects, giving an observed defect rate of (11/200=0.055), or 5.5%. For the initial exercise, use the assumed model parameter (p=0.05).

Batch	Products inspected	Defective products observed
1	20	1
2	20	0
3	20	2
4	20	1
5	20	3
6	20	0
7	20	1
8	20	2
9	20	1
10	20	0

Which probability models apply?

A. Binomial distribution — defects per batch

Question: In a batch of 20 products, what is the probability of finding exactly 2 defective products?

$$X \sim \text{Binomial}(20, 0.05)$$

$$P(X = 2) \approx 0.1887$$

The expected number of defects is $np = 1$ per batch.

B. Poisson distribution — defect counts

Question: If the average is 5 defects per 100 products, what is the probability of observing 3 defects in a batch of 100?

$$X \sim \text{Poisson}(5)$$

$$P(X = 3) \approx 0.1404$$

This is a Poisson approximation to the Binomial model when events are relatively rare.

C. Geometric distribution — inspections until a defect

Question: If each inspected product independently has a 5% defect probability, how many inspections might be needed before finding the first defective product?

$$X \sim \text{Geometric}(0.05)$$

$$E[X] = 1/0.05 = 20$$

The expected waiting time is 20 inspections.

These are three different views of the same quality-control process: defects in a fixed batch, defect counts in a production interval, and the waiting time until the first defect.

Steps to experiment on web application

1. Open the **Binomial** module in your Web application.
2. Enter `n = 20` and `p = 0.05`.
3. Calculate the mean, variance, and probability of exactly 2 defects.
4. View the PMF graph. The horizontal axis represents the number of defects and the vertical axis represents the probability of each count.
5. Change the defect probability from `0.05` to `0.10`. Observe how the distribution shifts toward larger defect counts.
6. Open the Poisson module and enter `λ = 5` for 100 inspected products. Compare the Poisson probabilities with the Binomial results.
7. Open the Geometric module and enter `p = 0.05`. Examine the probability of detecting the first defect on different inspection attempts.
8. Write down the numerical results, graph observations, and an inference about the quality-control process.

Use case 2: Cloud operations and cybersecurity monitoring

Industry problem statement

A cloud service provider operates servers and APIs used by thousands of customers. Its operations team monitors failed requests, security alerts, and retry attempts.

- The team wants to understand:
  - How many requests are likely to fail in a fixed set of requests?
  - How many security alerts might occur during a fixed time interval?
  - How many retry attempts might be required before a request succeeds?
  - These questions connect naturally to Binomial, Poisson, and Geometric distributions.

Sample dataset: 12 monitoring intervals

- The following is an illustrative dataset. Each interval is 10 minutes long.

Interval	API requests	Failed requests	Security alerts
1	100	2	2
2	100	1	4
3	100	3	3
4	100	2	1
5	100	4	5
6	100	1	2
7	100	2	4
8	100	3	3
9	100	2	2
10	100	1	6
11	100	4	1
12	100	2	3
Total	**1,200**	**27**	**36**

From the data:

- Observed API failure rate = $27/1200 = 0.0225$, or **2.25%**.
- Average security alerts per 10-minute interval = $36/12 = 3$.

These values can be used to initialize the Web application.

Apply the three probability models

A. Binomial — API request failures

Assume a request has a constant failure probability of $p = 0.0225$. For 100 independent requests:

$$X \sim \text{Binomial}(100, 0.0225)$$

Expected failures:

$$E[X] = 100(0.0225) = 2.25$$

Ask: What is the probability of exactly 3 failed requests in the next 100 requests?

B. Poisson — security alerts

The dataset has an average of 3 alerts per 10 minutes:

$$X \sim \text{Poisson}(3)$$

$$P(X = 5) \approx 0.1008$$

Ask: What is the probability of observing 5 security alerts in the next 10-minute interval?

C. Geometric — retry attempts

Suppose a request succeeds independently on each attempt with probability $p = 0.8$.

$$X \sim \text{Geometric}(0.8)$$

$$E[X] = 1/0.8 = 1.25$$

Ask: What is the probability that the first successful request occurs on the third attempt?

$$P(X = 3) = 0.2^2(0.8) = 0.032$$

Therefore, the probability is **3.2%**.

Note that the retry probability is an assumed parameter for this exercise; it is not estimated from the supplied monitoring table. Real retry attempts may be dependent, particularly when the server is experiencing an outage.

Steps to experiment on web application

1. **Enter the data:** Use the monitoring table to identify the failure rate and average alert count.
2. **Select Binomial:** Set $n = 100$ and $p = 0.0225$. Calculate the probability of exactly 3 failures and view the PMF.
3. **Select Poisson:** Set $\lambda = 3$. Calculate the probabilities of 0, 1, 2, 3, and higher alert counts. View the graph.
4. **Experiment with λ:** Compare $\lambda = 2$, $\lambda = 3$, and $\lambda = 5$. Observe how the distribution shifts as the average alert count increases.
5. **Select Geometric:** Set $p = 0.8$. Calculate the probability of success on the first, second, and third attempts.
6. **Interpret the results:** Discuss whether the calculated probabilities could help the operations team plan monitoring thresholds, investigate failures, or evaluate retry policies.